

CUTANEOUS DISORDERS OCCURRING IN HUMAN IMMUNODEFICIENCY VIRUS DISEASE

Pruritus and Pruritic Eruptions of Human Immunodeficiency Virus Disease

Pruritus is a common complaint in patients with late symptomatic and advanced HIV disease and serves as a surrogate cutaneous marker for disease progression. It occurs more frequently in individuals with CD4 counts below 50 cells/ μ L compared to those with counts above 250 cells/ μ L. In most cases, pruritus results from primary or secondary dermatoses rather than metabolic disorders.

Differential Diagnosis of Primary Pruritic Skin Disorders

The differential diagnosis of pruritic skin conditions in HIV-infected individuals includes: - Eosinophilic folliculitis (EF) - Papular pruritic eruption of HIV (PPE) - Adverse cutaneous drug eruptions - Atopic dermatitis - Xerosis - Dermographism - Allergic contact dermatitis - Scabies - Insect bites

Less commonly, systemic and metabolic disorders such as lymphoma, renal failure, viral hepatitis (hepatitis B or C), and obstructive liver disease can cause pruritus in the absence of visible skin lesions—termed "metabolic" pruritus.

An atopic diathesis may emerge in individuals with advanced HIV disease and pruritus. Chronic rubbing and scratching can lead to secondary changes such as excoriations, atopic dermatitis, lichen simplex chronicus, and prurigo nodularis.

Secondary *Staphylococcus aureus* infection—manifesting as impetiginization, furunculosis, or cellulitis—is very common in these traumatized areas.

One cross-sectional study found that, in addition to folliculitis, prurigo is the most common cutaneous disorder in patients with CD4⁺ counts below 200 cells/ μ L. This finding remained consistent even among individuals who had never received HAART. Conversely, idiopathic pruritus and candidiasis were most prevalent in those with viral loads exceeding 55,000 copies/ μ L. These observations suggest that certain dermatoses are linked to low CD4⁺ counts, while others may result from direct viral effects.

Ichthyosis vulgaris and xerosis are frequent in advanced HIV disease and may be associated with mild pruritus. Protease inhibitors (PIs), particularly indinavir, often cause xerosis, sometimes accompanied by eczematous or nummular dermatitis, which may lead to mild or moderate pruritus. These drug-related changes typically appear soon after initiating therapy.

Eosinophilic Folliculitis and Papular Pruritic Eruption of Human Immunodeficiency Virus

Eosinophilic Folliculitis (EF)

EF is a chronic, pruritic dermatosis seen in individuals with advanced HIV disease. Studies have shown an association between EF and low CD4 counts (typically below 125 cells/ μ L) and low CD4 nadir (average around 66 cells/ μ L), regardless of antiretroviral therapy status. EF can also occur during immune reconstitution inflammatory syndrome (IRIS).

The exact etiology and pathogenesis of EF remain unclear, but it may involve a T helper 2 (Th2)-mediated immune response to an unknown antigen.

Clinical Features

- Small, pink to red, edematous, folliculocentric papules
- Less commonly, pustules may be present

- Lesions are distributed symmetrically above the nipple line, affecting the chest, proximal arms, head, and neck